

Topical Outline

Project Title:

LeaveSync: A Leave Management System with Device Fingerprinting and Digital Signature-Based Approval for The Lewis College

A. Conceptual Foundations

- Leave Management Systems Overview
- Human Resource Management Systems (HRMS) Concepts
- Workflow Automation in Administrative Processes
- Information Security Principles in Information Systems
- Authentication and Authorization Concepts
- Data Privacy and Protection in Organizational Systems

B. Secure Leave Management Concepts

- Digital Leave Application and Approval Workflow
- Device Fingerprinting for User and Device Identification
- Trusted Device Registration and Access Control
- Digital Signature-Based Approval and Verification
- Non-Repudiation and Data Integrity in Transactions
- Role-Based Access (Admin, HR, Employee) in Leave Systems

C. System Technologies

- Web-Based System Development (HTML, CSS, JavaScript)
- Backend Development (PHP / Node.js / Python)
- Database Management Systems (MySQL / PostgreSQL)
- Device Fingerprinting Technologies (Browser, OS, IP-based Identification)
- Digital Signature Technologies (Cryptographic Signing, WebAuthn / PKI Concepts)
- Secure Communication Protocols (HTTPS, Encryption)

D. System Design and Architecture

- Client-Server Architecture for Web-Based Systems
- System Modules (User, HR, Admin)
- Leave Processing and Approval Workflow Design
- Device Fingerprinting Integration in System Access
- Digital Signature Integration in Approval Process
- Database Design for Leave Records and User Data

E. Features and Functional Components

- Device-Aware Leave Application Submission
- Secure Login and Trusted Device Verification
- Digital Signature-Based Leave Approval
- Automated Leave Tracking and Balance Monitoring
- Notification System for Leave Status Updates
- Leave History and Report Generation

F. System Evaluation and Impact

- System Usability and User Satisfaction
- Efficiency in Leave Processing and Approval Time
- Accuracy of Leave Records and Data Management
- Effectiveness of Device Fingerprinting in Security
- Reliability of Digital Signatures for Approval Integrity
- Impact on HR Workload and Organizational Productivity